Chapter 27: Wave Optics

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PHY201

Lecture 27

Part I

Wave Nature of Light

Outline of Part I: Wave Nature of Light

Outline of Part I: Wave Nature of Light

Introduction

The Wave Aspect of Light

Huygens' Principle

Wave Optics: The Limits of Ray Optics

Ray optics explains reflection, refraction, and image formation — but it fails whenever light interacts with structures *comparable in size to its wavelength.* Wave optics explains what ray optics cannot.

Property	**Ray Optics**	**Wave Optics**
Model of light	Straight ray	Wave with phase
Key quantity	Direction	Phase difference
Predicts	Images, shadows	Interference, diffraction
Breaks down when	Size $\sim \lambda$	Never (more general)

Unifying Idea

Every wave-optics phenomenon in this chapter arises from two principles: (1) waves superpose, and (2) relative phase determines what we observe.

Chapter Roadmap

All Topics Connect Through Phase Difference

- **27.1** Wavelength in media — phase accumulates differently in different materials
- **27.2** Huygens' principle — explains *why* waves bend around obstacles
- **27.3** Young's double slit — cleanest demonstration of interference
- **27.4** Diffraction gratings — interference from many slits
- **27.5** Single-slit diffraction — apertures create patterns
- **27.6** Rayleigh criterion — diffraction limits resolution
- **27.7** Thin films — phase shifts from reflection
- **27.8** Polarization — reveals transverse nature of light

Outline of Part I: Wave Nature of Light

Wavelength and Frequency in a Medium

Light in a Medium of Index n

Speed is reduced: $v = c/n$.

Frequency is set by the source — **it does not change** at an interface.

Since $v = f\lambda$, the wavelength must decrease:

$$\boxed{\lambda_n = \frac{\lambda}{n}}$$

where λ is the vacuum wavelength.

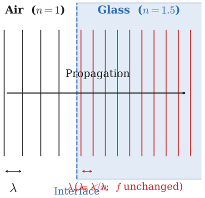
Common Misconception

Light slows in glass, therefore frequency decreases.

Wrong. Frequency is fixed by the source:

Intuition

Boundary conditions require the wave frequency to be continuous across any interface. The medium cannot "create or destroy" oscillation cycles.



Check Your Understanding

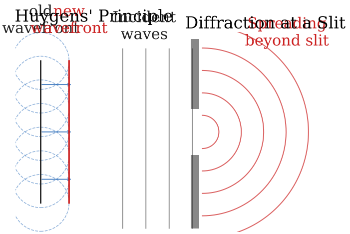
A red laser ($\lambda = 633\,\text{nm}$ in air) enters glass with $n = 1.50$. Which of the following **change** inside the glass?

(A) Frequency (B) Wavelength (C) Speed (D) Both B and C

Come to lecture to see the answer.

Outline of Part I: Wave Nature of Light

Huygens' Principle



Huygens' Principle

Every point on a wavefront acts as a **point source** of new spherical wavelets.

The new wavefront is the common tangent (envelope) to all these secondary wavelets.

Why Diffraction Happens

At a narrow opening, only a few secondary sources exist. Their wavelets spread in all directions — bending around the edges.

Diffraction is largest when opening $\sim \lambda$

Diffraction and Aperture Size

The amount of diffraction depends on the ratio of wavelength to aperture size.

Wide opening $D \gg \lambda$	Moderate opening $D \sim \lambda$	Tiny opening $D \ll \lambda$
Straight-through beam; negligible spreading. Ray optics is adequate.	Significant bending at edges. Diffraction pattern visible.	Nearly hemispherical spreading. Ray optics completely fails.

Key Relationship

$$\text{Smaller aperture} \longrightarrow \text{more diffraction} \longrightarrow \text{wider spread}$$

Part II

Interference

Outline of Part II: Interference

Young's Double-Slit Experiment

Diffraction Gratings

Outline of Part II: Interference

Young's Double-Slit Experiment (1801)

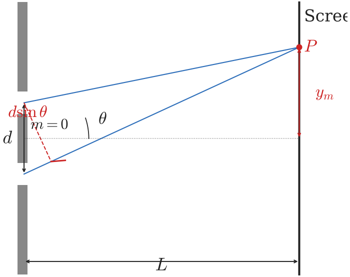
Thomas Young demonstrated that light is a wave by showing that two coherent light sources produce a stable **interference pattern** — alternating bright and dark bands called **fringes**.

- Light passes through two narrow slits separated by distance d
- Each slit becomes a coherent secondary source (Huygens)
- The two sets of waves overlap and interfere on a screen at distance L
- Bright bands appear where waves **add** (constructive)
- Dark bands appear where waves **cancel** (destructive)

Historical Significance

Ray optics predicts *two bright stripes*. Wave optics predicts a *fringe pattern*. Young's experiment decided the debate in favor of waves.

Double-Slit Geometry and Path Difference



Path Difference

The extra distance traveled by the wave from the farther slit is:

$$\Delta\ell = d \sin\theta$$

Interference Condition

Constructive (bright): path difference $= m\lambda$

Destructive (dark): path difference $= (m + \frac{1}{2})\lambda$

Double-Slit Equations

Constructive (Bright Fringes)

$$\boxed{d\sin\theta = m\lambda} \qquad m = 0, \pm 1, \pm 2, \ldots$$

$m = 0$ is the central maximum

Destructive (Dark Fringes)

$$\boxed{d\sin\theta = \left(m + \tfrac{1}{2}\right)\lambda}$$

Small-Angle: $y_m = m\lambda L/d$

Spacing: $\Delta y = \lambda L/d$

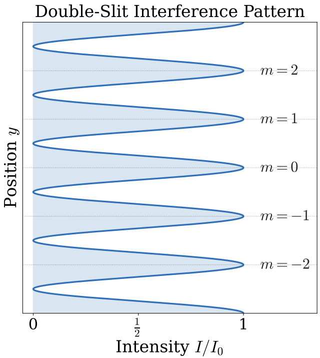
Conceptual Chain

- Path diff $\Delta\ell = d\sin\theta$
- $\rightarrow$ Phase difference
- $\rightarrow$ Amplitude
- $\rightarrow$ Intensity $\propto A^2$

Energy

Redistributed, not destroyed at dark fringes.

The Interference Pattern



Pattern Features

- Equally spaced bright fringes
- Spacing $\Delta y = \lambda L/d$ increases with λ or L
- Spacing decreases as d increases
- Pattern is *wider* for smaller slit separation

Energy Is Redistributed

Dark fringes do not mean "no light arrived." Energy is **redistributed** from dark regions to bright regions. Total energy is conserved.

Example: Fringe Spacing — (cp2e_ch27_ex1)

Calculating Fringe Spacing

In a double-slit experiment, the slits are 0.0100 mm apart and the screen is 2.00 m away. The light has $\lambda = 633\,\text{nm}$. What is the spacing between bright fringes?

$$\Delta y = \frac{\lambda L}{d}$$

$$\Delta y = \frac{(633 \times 10^{-9}\,\text{m})(2.00\,\text{m})}{1.00 \times 10^{-5}\,\text{m}}$$

$$\boxed{\Delta y = 0.127\,\text{m} = 12.7\,\text{cm}}$$

Fringes are well-separated — easily visible at 2 m. Note that $d = 10\,\mu\text{m}$ is only about 16 wavelengths wide.

- What happens to Δy if d doubles?
- What happens if λ increases?

Example: Finding Wavelength — (cp2e_ch27_ex2)

Wavelength from Double-Slit Pattern

Bright fringes in a double-slit experiment are separated by 0.0400 m. The slits are 5.00×10^{-5} m apart and the screen is 3.00 m away. Find the wavelength of light used.

$$\Delta y = \frac{\lambda L}{d}$$

$$\lambda = \frac{\Delta y \cdot d}{L}$$

$$\lambda = \frac{(0.0400\,\text{m})(5.00 \times 10^{-5}\,\text{m})}{3.00\,\text{m}}$$

$$\boxed{\lambda = 6.67 \times 10^{-7}\,\text{m} = 667\,\text{nm}}$$

667 nm is in the red portion of the visible spectrum (620 nm–750 nm).

- Can we determine wavelength more precisely by measuring multiple fringe spacings? Why?

Check Your Understanding

In a double-slit experiment, the slit separation d is **doubled** while all other quantities stay the same. What happens to the fringe spacing Δy?

Come to lecture to see the answer.

Outline of Part II: Interference

Young's Double-Slit Experiment

Diffraction Gratings

Multiple-Slit Diffraction: The Diffraction Grating

A **diffraction grating** contains hundreds to thousands of slits per millimeter. The result is the *same interference condition* as the double slit, but with much **sharper, brighter maxima**.

Grating Equation (same form as double slit)

$$\boxed{d \sin\theta = m\lambda}, \qquad m = 0, \pm 1, \pm 2, \ldots$$

d = grating spacing (distance between adjacent slits)

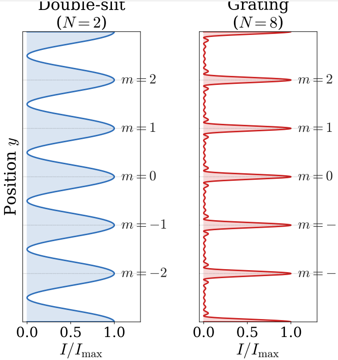
Why Sharper?

With many slits, destructive interference is *more complete* between maxima, so the bright lines become much narrower.

More slits = sharper lines = better wavelength resolution.

If the grating has N slits/mm, then $d = 1/N$ (in mm)

Diffraction Grating Pattern



Pattern Features

- Same peak *positions* as double slit ($d \sin\theta = m\lambda$)
- Peaks are *much narrower*
- Different wavelengths diffract to different angles
- Gratings **separate colors** — used in spectroscopy

Application: Spectroscopy

Diffraction gratings measure wavelength to high precision in astronomy, chemistry, and materials science.

Example: Diffraction Grating — (cp2e_ch27_ex3)

Grating Angle for First-Order Maximum

A diffraction grating has 600 lines/mm. At what angle does the first-order maximum appear for light with $\lambda = 550\,\text{nm}$?

$$d = \frac{1}{600\,\frac{1}{\text{mm}}} = 1.67 \times 10^{-3}\,\text{mm}$$

$$= 1.67 \times 10^{-6}\,\text{m}$$

$$d \sin\theta = m\lambda$$

$$\sin\theta = \frac{m\lambda}{d} = \frac{(1)(550 \times 10^{-9}\,\text{m})}{1.67 \times 10^{-6}\,\text{m}}$$

$$\boxed{\theta = \arcsin(0.329) = 19.2^\circ}$$

First-order means $m = 1$. The $m = -1$ maximum appears symmetrically at $\theta = -19.2^\circ$.

- Would a *finer* grating (more lines/mm) give a larger or smaller θ for the same λ?

Check Your Understanding

A diffraction grating is illuminated with white light. The $m = 1$ maximum for red light ($\lambda \approx 700\,\text{nm}$) appears at a **larger or smaller** angle than the $m = 1$ maximum for violet light ($\lambda \approx 400\,\text{nm}$)?

Come to lecture to see the answer.

Part III

Diffraction

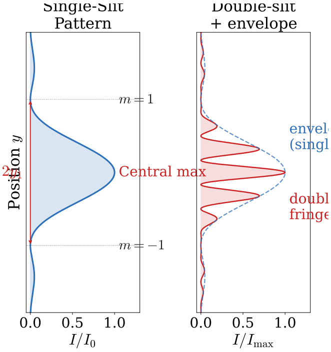
Outline of Part III: Diffraction

Outline of Part III: Diffraction

Single-Slit Diffraction

Single-Slit Diffraction

Even a *single* slit produces a diffraction pattern. The slit width D acts as many secondary Huygens sources that interfere with each other.



Destructive Interference Minima

$$\boxed{D \sin\theta = m\lambda}, \quad m = \pm 1, \pm 2, \ldots$$

$D =$ slit width, $m \neq 0$

The central maximum spans from $-\theta_1$ to $+\theta_1$ — twice as wide as any side lobe.

Single-Slit Pattern Features

Width of Central Maximum

Angular half-width of central maximum:

$$\theta_1 = \arcsin\left(\frac{\lambda}{D}\right) \approx \frac{\lambda}{D}$$

Linear half-width on screen at distance L:

$$y_1 \approx \frac{\lambda L}{D}$$

Opposite to Double-Slit

Narrower slit $\Rightarrow$ *wider* central maximum. ($D \downarrow$ means $\theta_1 \uparrow$)

Pattern Summary

- Central max: bright, wide, contains ~84% of light
- Side maxima: weaker, narrower
- Dark minima at $D \sin\theta = m\lambda$
- Wider slit $\Rightarrow$ narrower central max

Example: Single-Slit Width — (cp2e_ch27_ex4)

Width of Central Diffraction Maximum

Light of wavelength 589 nm passes through a slit 0.250 mm wide and falls on a screen 1.00 m away. Find the half-width of the central maximum.

$$D \sin\theta_1 = (1)\lambda$$

$$\sin\theta_1 = \frac{\lambda}{D} = \frac{589 \times 10^{-9}\,\mathrm{m}}{2.50 \times 10^{-4}\,\mathrm{m}}$$

$$= 2.356 \times 10^{-3}$$

$$y_1 = L\tan\theta_1 \approx L\sin\theta_1$$

$$\boxed{y_1 = (1.00)(2.356 \times 10^{-3}) = 2.36\,\mathrm{mm}}$$

The full central maximum width is $2y_1 = 4.72\,\mathrm{mm}$.

The slit is only 0.250 mm wide, yet the central maximum spreads to nearly 5 mm — diffraction has widened the beam by a factor of ~ 19.

- If D doubles, what happens to y_1?

Single Slit Envelope of Double-Slit Pattern

The Real Double-Slit Pattern

Real slits have finite width D as well as separation d. The result:

- Double-slit fringes appear *inside* the single-slit envelope
- Some double-slit maxima are *missing* where they coincide with single-slit minima

Connection: One Slit to Many

- 1 slit: broad single-slit envelope
- 2 slits: narrow fringes inside envelope
- Grating (N slits): razor-sharp peaks at same positions

All are manifestations of the *same* physics: path differences and superposition. More slits simply sharpen the constructive-interference peaks.

Check Your Understanding

A single slit is illuminated with green light ($\lambda = 520\,\text{nm}$) and produces a central maximum of half-width $3.0\,\text{mm}$ on a screen $1.5\,\text{m}$ away. If the same slit is illuminated with red light ($\lambda = 700\,\text{nm}$), will the central maximum be **wider, narrower, or the same width**?

Come to lecture to see the answer.

Part IV

Resolution and Thin Films

Outline of Part IV: Resolution and Thin Films

Outline of Part IV: Resolution and Thin Films

Limits of Resolution

Thin-Film Interference

Why Resolution Has a Limit

Any opening — a lens, a telescope aperture, a microscope objective — diffracts light. This creates a minimum spot size for any focused image, which **limits how closely spaced two objects can be and still be distinguished**.

Diffraction from a Circular Aperture

A circular aperture of diameter D produces a central disk (Airy disk) surrounded by rings. The angular radius of the first dark ring is:

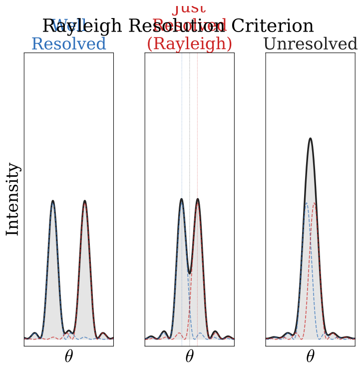
$$\boxed{\theta = 1.22\,\frac{\lambda}{D}}$$

Rayleigh Criterion

Two sources are **just resolved** when the central maximum of one falls on the first minimum of the other.

Minimum resolvable angle: $\theta_{\text{min}} = 1.22\,\lambda/D$

Rayleigh Criterion Illustrated



Three Cases

- **Resolved:** two distinct peaks visible
- **Just resolved:** central max of one at first min of other (Rayleigh)
- **Unresolved:** peaks merge into one blob

To Improve Resolution

Use shorter λ (blue or UV light) or larger aperture D.

Example: Telescope Resolution — (cp2e_ch27_ex5)

Minimum Resolvable Angle of a Telescope

The Hubble Space Telescope has a primary mirror diameter of $D = 2.40\,\text{m}$. What is its minimum resolvable angle for light with $\lambda = 550\,\text{nm}$?

$$\theta_{\text{min}} = 1.22\,\frac{\lambda}{D}$$

$$\theta_{\text{min}} = 1.22\,\frac{550 \times 10^{-9}\,\text{m}}{2.40\,\text{m}}$$

$$\boxed{\theta_{\text{min}} = 2.80 \times 10^{-7}\,\text{rad} \approx 0.058''}$$

This is about 10 times better than the best ground-based telescope (limited by atmospheric turbulence, not diffraction).

Larger mirror $\Rightarrow$ smaller θ_{min} $\Rightarrow$ finer detail.

- Why does the factor 1.22 appear? (Circular vs. rectangular aperture)

Example: Microscope Resolution — (cp2e_ch27_ex6)

Limit of a Light Microscope

A light microscope uses $\lambda = 400\,\text{nm}$ (violet) and has a lens aperture of $D = 0.800\,\text{mm}$ with the object at $1.00\,\text{mm}$ from the lens. What is the minimum resolvable separation of two points?

$$\theta_{\min} = 1.22\,\frac{\lambda}{D}$$

$$= 1.22\,\frac{400 \times 10^{-9}\,\text{m}}{8.00 \times 10^{-4}\,\text{m}}$$

$$= 6.10 \times 10^{-4}\,\text{rad}$$

$$s = L\,\theta_{\min} = (1.00 \times 10^{-3}\,\text{m})(6.10 \times 10^{-4})$$

$$\boxed{s = 0.610\,\mu\text{m}}$$

About $0.6\,\mu$m — comparable to the size of bacteria.

This is the fundamental diffraction limit of a visible-light microscope. To see smaller, use shorter λ (electron microscope uses $\lambda \sim$ pm).

- Why can't we just use a higher magnification to see smaller objects?

Check Your Understanding

You want to improve the resolution of a telescope. Rank these strategies from **most to least effective**, assuming equal cost:

(A) Double the aperture diameter (B) Halve the wavelength of light used (C) Both A and B

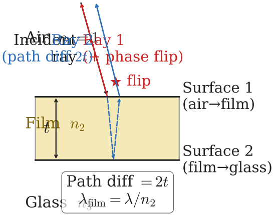
Come to lecture to see the answer.

Outline of Part IV: Resolution and Thin Films

Limits of Resolution

Thin-Film Interference

Thin-Film Interference: The Setup



Two Reflected Rays

- Ray 1: reflects off top surface
- Ray 2: transmits into film, reflects off bottom, exits
- Path difference: $2t$ (film thickness t, at normal incidence)

Phase Flip Rule

Reflection from a *higher*-n medium adds $\lambda/2$ phase shift (like a string reflecting off a wall).

Reflection from a *lower*-n medium adds no phase shift.

Thin-Film Decision Tree

Step-by-Step Procedure

1. **Count phase flips.** Each reflection from higher-n medium contributes a $\lambda/2$ shift.
2. **Determine path difference.** At near-normal incidence: $\Delta\ell = 2t$. Use $\lambda_n = \lambda/n$ inside the film.
3. **Combine.** If #flips is *odd*, the phase flip adds an effective extra $\lambda_n/2$.
4. **Apply condition.** Constructive if total effective path difference $= m\lambda_n$; destructive if $= (m + \frac{1}{2})\lambda_n$.

# Flips	Constructive	Destructive
0 or 2 (even)	$2t = m\lambda_n$	$2t = (m + \frac{1}{2})\lambda_n$
1 (odd)	$2t = (m + \frac{1}{2})\lambda_n$	$2t = m\lambda_n$

Example: Soap Bubble — (cp2e_ch27_ex7)

Constructive Interference in a Soap Film

A soap film ($n = 1.33$) in air has thickness t. For what minimum thickness does it appear bright (constructive) in green light ($\lambda = 532\,\text{nm}$)?

Air(n=1) →Soap(1.33) →Air(1): **1 flip**

$$1 \text{ flip} \Rightarrow 2t = \left(m + \tfrac{1}{2}\right) \lambda_n$$

$$\lambda_n = \frac{\lambda}{n} = \frac{532\,\text{nm}}{1.33} = 400\,\text{nm}$$

$$2t = \left(0 + \tfrac{1}{2}\right) (400\,\text{nm})$$

$$\boxed{t_{\text{min}} = 100\,\text{nm}}$$

Only one phase flip (air → soap), so constructive needs the half-integer form.

$t = 100\,\text{nm}$ — much thinner than a human hair ($\sim 70\,\mu\text{m}$).

- Why does a soap bubble appear black just before it pops?

Example: Anti-Reflection Coating — (cp2e_ch27_ex8)

Minimum Thickness for Non-Reflective Coating

A glass lens ($n_g = 1.70$) is coated with MgF_2 ($n_c = 1.38$). What minimum coating thickness eliminates reflection of $\lambda = 550\,\text{nm}$ light?

Air(1) $\rightarrow$MgF$_2$(1.38) $\rightarrow$Glass(1.70): **2 flips**

2 flips $\Rightarrow$ destructive: $2t = m\lambda_n$

$$\lambda_n = \frac{550\,\text{nm}}{1.38} = 398\,\text{nm}$$

$$2t = (1)(398\,\text{nm})$$

$$\boxed{t_{\text{min}} = 99.6\,\text{nm}}$$

Two flips give net zero phase shift, so *destructive* needs the integer form (not half-integer).

Camera lenses and eyeglasses use coatings like this to minimize reflections.

- If the coating is for green light, what color would you see reflected?

Check Your Understanding

A glass slide ($n = 1.52$) has a thin film of oil ($n = 1.45$) on its top surface, with air above the oil. How many phase flips occur for light reflected from (a) the air–oil interface and (b) the oil–glass interface?

Come to lecture to see the answer.

Part V

Polarization

Outline of Part V: Polarization

Outline of Part V: Polarization

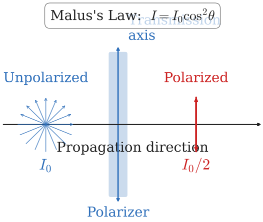
Polarization

Microscopy and Wave Optics

Polarization: Light Is a Transverse Wave

Polarization is possible *only* for transverse waves. The fact that light can be polarized is direct evidence that light is a transverse wave (not a longitudinal pressure wave like sound).

- Unpolarized light: $\vec{E}$ oscillates in all directions perpendicular to the propagation direction
- Linearly polarized light: $\vec{E}$ oscillates in only one direction
- A **polarizer** transmits only the component of $\vec{E}$ along its transmission axis

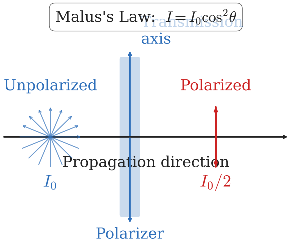


Malus's Law

Polarized I_0 through analyzer at angle θ:

$$\boxed{I = I_0 \cos^2 \theta}$$

- $\theta = 0$: $I = I_0$ (full transmission)
- $\theta = 90°$: $I = 0$ (crossed — dark)
- Unpolarized → one polarizer: $I = \frac{1}{2} I_0$



Example: Malus's Law — (cp2e_ch27_ex9)

Two Polarizers

Unpolarized light of intensity $10.0\,\frac{\mathrm{W}}{\mathrm{m}^2}$ passes through two polarizers. The second polarizer's axis is 30.0° from the first. What intensity emerges from the second polarizer?

After 1st polarizer:

$$I_1 = \frac{1}{2}\,I_0 = \frac{1}{2}(10.0\,\frac{\mathrm{W}}{\mathrm{m}^2})$$
$$= 5.00\,\frac{\mathrm{W}}{\mathrm{m}^2}$$

Two steps: first polarizer halves intensity; second applies Malus's law to the now-polarized beam.

After 2nd (Malus):

$$I_2 = I_1\cos^2(30.0°)$$
$$I_2 = (5.00\,\frac{\mathrm{W}}{\mathrm{m}^2})(0.866)^2$$
$$\boxed{I_2 = 3.75\,\frac{\mathrm{W}}{\mathrm{m}^2}}$$

- What if a *third* polarizer at 45° from the second is added?
- Can adding a third polarizer *increase* the transmitted intensity compared to two crossed polarizers?

Brewster's Angle

Brewster's Law

Reflection at angle θ_b: reflected ray is *fully* polarized.

$$\boxed{\tan\theta_b = \frac{n_2}{n_1}}$$

The reflected and refracted rays are perpendicular.

Application: Polarized Sunglasses

Glare from flat surfaces is partially polarized. Polarized sunglasses block this horizontal component.

Example

Air (n_1=1.00) to glass (n_2=1.55):

$$\theta_b = \arctan\left(\frac{1.55}{1.00}\right) = 57.2^\circ$$

Other Polarization Mechanisms

Polarization can arise from several mechanisms, all exploiting the transverse nature of light.

- **Absorption polarizers (Polaroid):** anisotropic material absorbs one polarization component
- **Reflection** at Brewster's angle: reflected beam fully polarized
- **Scattering:** sky appears blue and polarized due to Rayleigh scattering by air molecules
- **Birefringence:** some crystals (calcite) have two different refractive indices and split light into two polarized beams
- **Liquid crystal displays (LCDs):** use tunable birefringence to control image brightness pixel by pixel

Check Your Understanding

Two crossed polarizers ($90°$) transmit no light. A **third polarizer** is inserted between them at $45°$ to each. Does light now emerge, and what fraction of I_0 passes through all three?

Come to lecture to see the answer.

Outline of Part V: Polarization

Microscopy Enhanced by Wave Optics

Modern microscopy techniques exploit every principle from this chapter.

- **Confocal microscopy:** point illumination reduces diffraction blur, improves resolution
- **Phase-contrast microscopy:** converts phase differences in transparent samples into visible intensity differences
- **Polarization microscopy:** reveals birefringent structures (crystals, muscle fibers) invisible in unpolarized light
- **STED microscopy:** uses interference to create a sub-diffraction excitation volume, breaking the Abbe limit
- **Electron microscopy:** uses electrons with $\lambda \sim 1\,\text{pm}$, improving resolution by $\sim 10^5$ over visible light

Synthesis

Chapter Summary: Wave Optics

Topic	Key Equation	Predicts
Wavelength in medium	$\lambda_n = \lambda/n$	Phase accumulation
Double slit (bright)	$d \sin\theta = m\lambda$	Bright fringe positions
Double slit (dark)	$d \sin\theta = (m + \frac{1}{2})\lambda$	Dark fringe positions
Single slit (dark)	$D \sin\theta = m\lambda$	Diffraction minima
Rayleigh criterion	$\theta = 1.22\,\lambda/D$	Angular resolution
Thin film	(depends on # phase flips)	Colors in reflection
Malus's law	$I = I_0 \cos^2\theta$	Polarizer output
Brewster's angle	$\tan\theta_b = n_2/n_1$	Reflected polarization

Unifying Principle

Superposition + phase difference $\longrightarrow$ all of wave optics.